PHYS 365 & PHYS 369
Midterm Exam

VERSION 01

October 28, 2025, 7:00-8:30 PM Time: 90 minutes

This is a closed-book exam containing a total of 25 equally weighted questions.

ONLY THE FOLLOWING ITEMS ARE ALLOWED ON YOUR DESK DURING THE EXAM:

1. THIS MULTIPLE-CHOICE QUESTION BOOKLET, which includes the exam questions, the formula sheet, and 3 blank pages. This booklet WILL **NOT** BE HANDED IN.

2. A BUBBLE SHEET used to answer multiple-choice questions. This will be handed in and it WILL BE GRADED.

3. A CALCULATOR. Only a Schulich-approved scientific calculator may be used during the exam.

4. A PEN OR PENCIL, used to black out your answers on the bubble sheet.

5. YOUR STUDENT ID CARD, used to check your identity during exam sign-in.

6. Students remaining after **65 minutes** must remain seated for the duration of the exam until permitted to leave by the exam supervisor.

1. When a wave hits a boundary between two media, which wave property must be the same for all 3 of the incident, reflected, and transmitted waves?

   A) The amplitude: $A_i = A_r = A_t$
   B) The frequency: $\omega_i = \omega_r = \omega_t$
   C) The wavenumber: $k_i = k_r = k_t$
   D) The wave speed: $v_i = v_r = v_t$

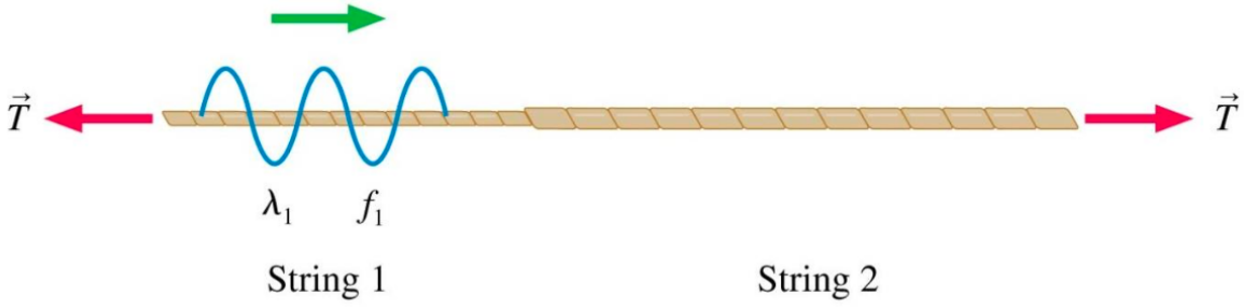
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2. For a standing wave on a string fixed at both ends, the fifth normal mode of vibration frequency corresponds to:

   A) One full wavelength fits into the string length.
   B) One and a half wavelengths fit into the string length.
   C) Two wavelengths fit into the string length.
   D) Two and a half wavelengths fit into the string length.

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3. A string under tension T with linear mass density μ_1 is joined to a second string with linear mass density μ_2 that is 9 times larger than μ_1. A sinusoidal wave carrying power P is sent along string 1. What is the power transmission coefficient for the transmitted wave?

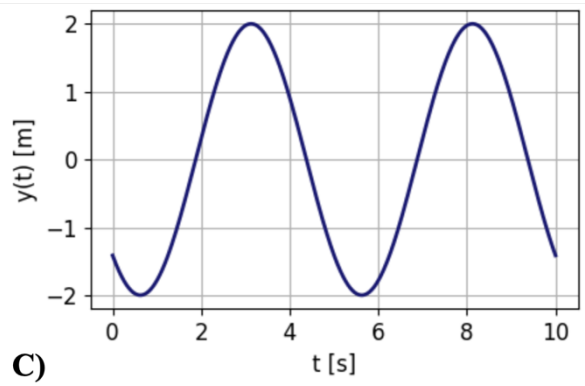


   A) $1/3$
   B) $2/3$
   C) $1/4$
   D) $3/4$

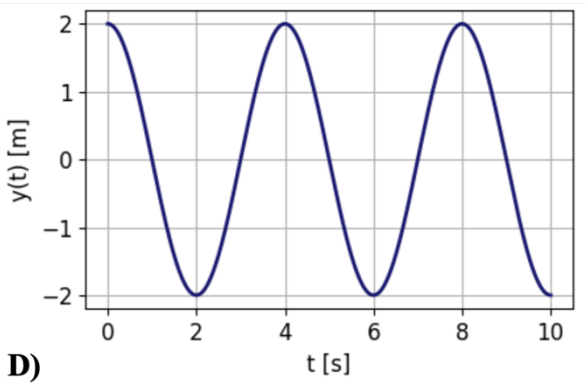
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4. Which of the following is a correct graph of a simple harmonic oscillator with an amplitude of 2 m, an angular frequency of $\frac{\pi}{2}$, and a phase constant of 0?

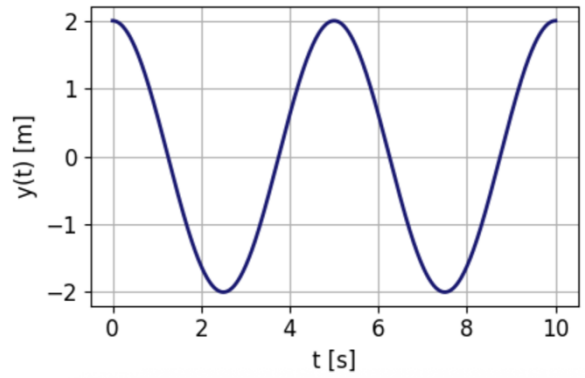
A)



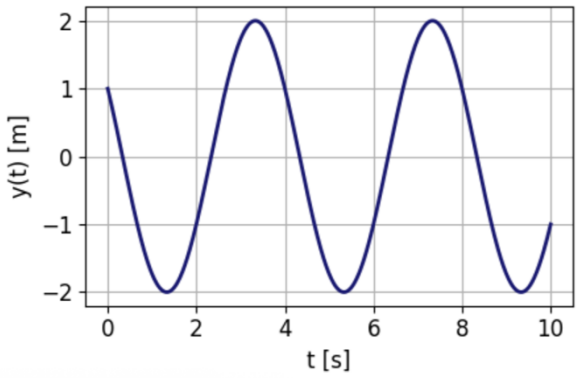
B)



C)

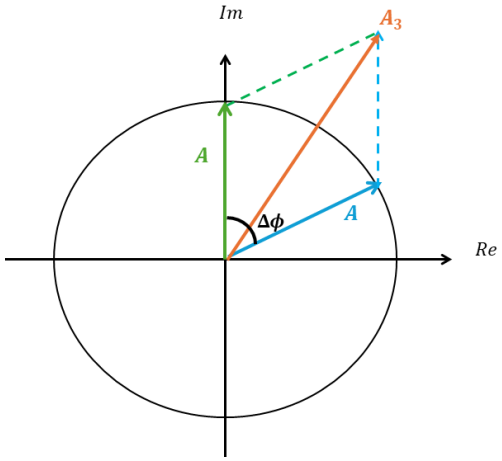


D)



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5. Two waves of equal amplitude A are traveling through the same medium with impedance Z_1. The picture shows the phasor representation of the two waves. Then the two waves cross a boundary into another medium with impedance $Z_2 > Z_1$. Which of the following statements regarding the phasor representation of the waves in the second medium is TRUE?



A) The angular frequency of the vectors spinning around the origin would stay the same, but the amplitudes of all the phasors would decrease.
B) The angular frequency of the vectors spinning around the origin and the amplitude of the resulting phasor would increase.
C) The amplitude of the initial phasors A and their phase difference $\Delta\phi$ would decrease such that the amplitude of the resulting phasor A_3 would stay the same.
D) The amplitudes and angular frequencies of all the phasors would remain unaffected by the new medium.

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6. A trombone has a variable length. When a musician blows into the mouthpiece and causes air in the tube of the horn to vibrate, the waves set up by the vibrations reflect back and forth in the horn to create standing waves. As the length of the horn is made shorter, what happens to the third harmonic frequency?

   A) The frequency remains the same.
   B) The frequency decreases.
   C) The frequency increases.
   D) The frequency increases or decreases depending on the diameter of the horn.

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7. A simple harmonic oscillator oscillates according to $y(t) = 1.2\cos\left(2\pi t + \frac{\pi}{6}\right)$. What is the complex amplitude?

   A) $1.2\, e^{\frac{\pi}{6}}$
   B) $e^{j\frac{\pi}{6}}$
   C) $1.2e^{j\frac{\pi}{6}+j\omega t}$
   D) $1.2e^{j\frac{\pi}{6}}$

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8. What is the relationship between the speed of sound in atomic helium gas He compared to the speed of sound in air? Assume that both are at the same temperature T and that the air is made from 80% nitrogen N_2 and 20% oxygen O_2.
   Mean atomic masses are: $M_{He} = 4.0026\ \text{g/mol}$, $M_{N_2} = 28.014\ \text{g/mol}$,
   $M_{O_2} = 32.00\ \text{g/mol}$.

   A) $v_{He} = 2.46\, v_{Air}$
   B) $v_{He} = 0.37\, v_{Air}$
   C) $v_{He} = 0.41\, v_{Air}$
   D) $v_{He} = 2.93 v_{Air}$

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9. Hydrogen atoms in the laboratory emit red light with wavelength 656 nm. When the light from a distant galaxy is viewed, this spectral line of hydrogen is observed to have a wavelength of 691 nm. What is the speed of this galaxy relative to the earth?

   A) 1.6×10^7 m/s
   B) 6.4×10^7 m/s
   C) 3.0×10^7 m/s
   D) 7.6×10^7 m/s

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10. A spring ($k = 50$ N/m) oscillates on a frictionless horizontal surface with a 5.0 kg mass attached to it. What is the angular frequency of the oscillation?

A) 0.1 rad/s
B) 10 rad/s
C) 3.2 rad/s
D) 5 rad/s

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11. A wave on a string under tension F is incident from the left on a boundary where the linear mass density doubles. What is the amplitude reflection coefficient?



A) 0.17
B) -0.17
C) 0.33
D) -0.33

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12. The low E and the high E strings on a guitar are the same length L, and are made from the same material. The high E string on a guitar vibrates with roughly 4 times the frequency of the low E string. Assuming both strings are under the same tension τ what is the relationship between the thicknesses/gauges of the two strings?
(Hint: Assume cylindrical strings. The wavelengths of the two waves are the same and are equal to twice the length of the string itself.)

A) The low E string is 2 times thicker than the high E string.
B) The low E string is 4 times thicker than the high E string.
C) The low E string is 8 times thicker than the high E string.
D) The low E string is 16 times thicker than the high E string.

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13. A mass of 1.5 kg is attached to a horizontal spring on a frictionless surface. At $t = 0$, the mass is released at $+50$ cm from equilibrium and accelerates at -42 m/s^2. What is the maximum speed of the mass?

A) 4.6 m/s
B) 42 m/s
C) 0.68 m/s
D) 458 m/s

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14. A pipe resonates in three successive normal modes at frequencies 431.25 Hz, 603.75 Hz and 776.25 Hz. The relevant speed of sound is 345 m/s. Based on this information, select the correct statement:

A) The pipe is open at both ends, and has length 1.0 m.
B) The pipe is open at one end, closed at the other, and has length 0.5 m.
C) The pipe is open at both ends, and has length 2.0 m.
D) The pipe is open at one end, closed at the other, and has length 1.0 m.

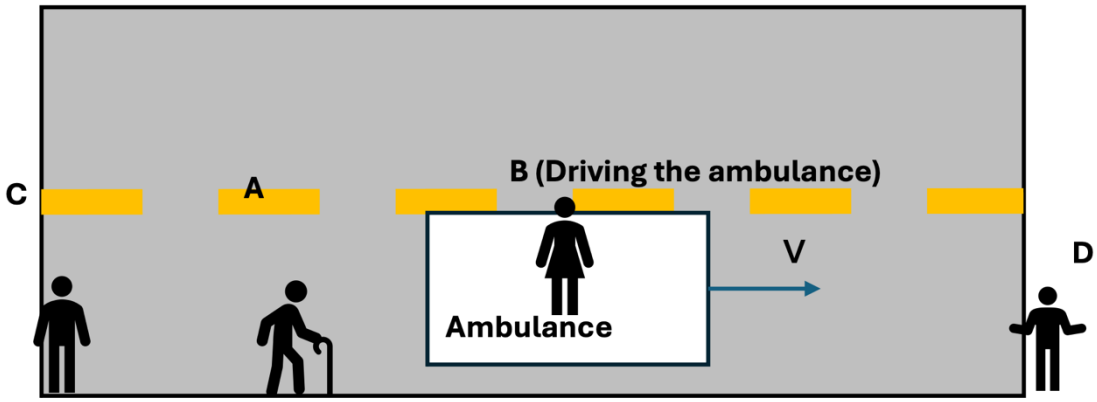
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15. Two AC currents enter a 3-wire junction. The first is a current of 1.4 A at a phase of 180°, and the second is 2.5 A at 60°. What is the phase shift of the resultant current?

A) 94.0º
B) –86.0º
C) 176º
D) 209º

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16. An ambulance turns on its siren as it is driving with speed v. At that time, four observers are positioned as per the diagram below. A, C, and D are stationary, and B is driving the ambulance. Order the observers from highest measured frequency to lowest measured frequency.



A) $f_D > f_B = f_A = f_C$
B) $f_D > f_B = f_A > f_C$
C) $f_D > f_B > f_A = f_C$
D) $f_D > f_B > f_A > f_C$

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17. A wave propagating on a string has the equation $y(x,t) = \frac{A}{2}\sin(kx - \omega t + \phi/2)$. If this wave is the result of the superposition of two initial waves with the same amplitude A, what is the phase difference ϕ between the two initial waves?

    A) $60.33°$
    B) $151.04°$
    C) $0°$
    D) $75.52°$

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18. Two monochromatic filters, one of which lets through light with wavelength $\lambda_1 = 650$ nm, are separately used in a Young's double slit experiment. The first (non-central) maximum of the interference pattern using filter 1 occurs in the same spot on the screen as the third maximum of the interference pattern using filter 2. What is the wavelength of light let through by the second filter?

    A) $\lambda_2 = 217$ nm
    B) $\lambda_2 = 1950$ nm
    C) $\lambda_2 = 1300$ nm
    D) $\lambda_2 = 325$ nm

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19. The clarinet is well-modeled as a cylindrical pipe that is open at one end and closed at the other. For a clarinet whose air column has an effective length of 66 cm, determine the frequency f_3 of the third normal mode of vibration. Use 340 m/s for the speed of sound inside the instrument.

    A) 193 Hz
    B) 386 Hz
    C) 579 Hz
    D) 773 Hz

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20. The pressure amplitude of a normal conversation sound wave is $\Delta p_m = 20$ mPa. The acoustic impedance of air is $Z_a = 412$ Pa · s/m^3. If we assume the frequency to be $f = 220$ Hz. What is the amplitude of the oscillation of the particles in the air?

    A) 35 nm
    B) 220 nm
    C) 220 µm
    D) 35 µm

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21. A wave travels according to the equation $y(x,t) = A\cos(kx - \omega t + \phi_0)$. If the frequency is increased while the wavenumber remains the same, which of the following is true?

A) The wave speed will increase.
B) The maximum particle velocity will increase.
C) Both the wave speed and maximum particle velocity will increase.
D) Neither the wave speed nor maximum particle velocity will increase.

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22. A speaker emits a sound wave that can be represented by the equation $\Delta P_1 = P_{max\,1}\cos(8.04x - 880\pi t)$. A second speaker right beside the first emits a sound wave at a slightly different frequency, resulting in a beat frequency of 6 Hz. Assuming the second speaker is emitting a higher frequency than the first speaker, which of the following equations could be used to describe this sound wave?

A) $\Delta P_2 = P_{max\,2}\cos(8.04x - 892\pi t)$
B) $\Delta P_2 = P_{max\,2}\cos(8.09x - 886\pi t)$
C) $\Delta P_2 = P_{max\,2}\cos(8.15x - 892\pi t)$
D) $\Delta P_2 = P_{max\,2}\cos(8.04x - 886\pi t)$

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23. I am shaking one end of a thick rope so there is a wave traveling along it. I am doing this exercise for 2 minutes. If I increase the amplitude of the wave by 22.5% and do the exercise again for 2 minutes, which is the TRUE statement?

A) The energy transported down the rope increases by 22.5%.
B) The energy transported down the rope increases by 50%.
C) The energy transported down the rope increases by 100%.
D) The energy transported down the rope increases by 122.5%.

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24. An ambulance with its siren on is approaching a car which is moving toward the ambulance. An observer in the car perceives the siren's frequency as:

A) lower than the actual frequency.
B) higher than the actual frequency.
C) the same as the actual frequency.
D) More information about the speed of the car and ambulance is needed.

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25. A coherent source of white light is used in a Young's double slit experiment. Select the TRUE statement.

A) Each maximum of the interference pattern will be a different color.
B) All maxima will be of white light.
C) Each maximum will be comprised of a rainbow of colors with the red end of the rainbow closest to the center.
D) The central maximum will be white, and each subsequent maximum will be comprised of a rainbow of colors with the blue end of the rainbow closest to the center.

PHYS 365 & 369 Formula Sheet

OSCILLATIONS

$x(t) = x_m \cos(\omega t + \phi)$	$\omega = 2\pi f = \frac{2\pi}{T}$	$\omega = \sqrt{\frac{k}{m}}$	$\omega = \sqrt{\frac{g}{\ell}}$	$T = 2\pi\sqrt{\frac{\ell}{g}}$

WAVES

$\frac{\partial^2 y}{\partial^2 t} = v^2 \frac{\partial^2 y}{\partial x^2}$	$y(x,t) = f(x \pm vt)$	$y(x,t) = y_m \cos(kx \pm \omega t + \phi)$	
$v = \frac{\omega}{k} = f\lambda$	$k = \frac{2\pi}{\lambda}$	$v = \frac{c}{n}$	$\lambda = \frac{\lambda_0}{n}$
$v = \sqrt{\frac{\tau}{\mu}}$	$v = \sqrt{\frac{\gamma RT}{M}}$	$v = \sqrt{\frac{Y}{\rho}}$	$v = \sqrt{\frac{B}{\rho}}$
$Z = \sqrt{\mu\,\tau} = \frac{\tau}{v} = \mu v$	$Z = \sqrt{\rho B} = \rho v$	$P_{avg} = \frac{1}{2}\mu v \omega^2 y_m^2$	$\Delta p_m = (v\rho\omega)s_m$
$\Delta p = \Delta p_m \sin(kx - \omega t)$	$I = \frac{1}{2}\rho v \omega^2 s_m^2$	$I = \frac{P}{4\pi R^2}$	$I = \frac{P}{A}$
$\beta = (10\ dB)\log_{10}\left(\frac{I}{I_0}\right)$	$\Delta\beta = (10\ dB)\log_{10}\left(\frac{I_2}{I_1}\right)$	$I(r_2) = \left(\frac{r_1}{r_2}\right)^{n-1} I(r_1)e^{-\alpha\Delta r}$	
$R = \frac{Z_1 - Z_2}{Z_1 + Z_2}$	$T = \frac{2Z_1}{Z_1 + Z_2}$	$B = -\frac{\Delta p}{\left(\frac{\Delta V}{V}\right)}$	$Y = \frac{\left(\frac{F}{A}\right)}{\left(\frac{\Delta L}{L}\right)}$
$R_e = R^2$	$T_e = \frac{Z_2}{Z_1}T^2$	$\Delta p = -B\frac{\partial s}{\partial x}$	

WAVE INTERFERENCE

$y'(x,t) = \left[2y_m \cos\left(\frac{1}{2}\phi\right)\right]\sin\left(kx - \omega t + \frac{1}{2}\phi\right)$		$y'(x,t) = [2y_m \sin(kx)]\cos(\omega t)$	
$\Delta\phi = \frac{\Delta L}{\lambda}2\pi$	$f_{beat} = f_1 - f_2$	$f' = f\frac{v \pm v_D}{v \mp v_S}$	$f' = f\sqrt{\frac{c \pm v}{c \mp v}}$

Constructive interference $\frac{\Delta L}{\lambda} = 0,\ 1,\ 2, \ldots$	Destructive interference $\frac{\Delta L}{\lambda} = 0.5,\ 1.5,\ 2.5, \ldots$
Taut string or pipe open at both ends $f_n = \frac{v}{\lambda_n} = \frac{nv}{2L}$ for $n = 1,\ 2,\ 3, \ldots$	Pipe one end closed, one end open $f_n = \frac{v}{\lambda_n} = \frac{nv}{4L}$ for $n = 1,\ 3,\ 5, \ldots$

$d\sin(\theta) = m\lambda$ for $m = 0, 1, 2, \ldots$ (maxima-bright fringes)

$d\sin(\theta) = \left(m + \frac{1}{2}\right)\lambda$ for $m = 0, 1, 2, \ldots$ (minima-dark fringes)

$2t = m\frac{\lambda_0}{n}, \quad m = 1, 2, 3 \ldots$		$2t = \left(m - \frac{1}{2}\right)\frac{\lambda_0}{n}, \quad m = 1, 2, 3 \ldots$	
Reflection	Reflection phase shift	$a\sin(\theta) = m\lambda$ for $m = 1, 2, 3, \ldots$ (minima)	
Off lower index	0		
Off higher index	0.5 wavelength		
$I = 4I_0\cos^2\left(\frac{1}{2}\phi\right)$	$\phi = \frac{2\pi d}{\lambda}\sin(\theta)$	$\theta_R = \sin^{-1}\left(\frac{1.22\lambda}{d}\right)$	
CONSTANTS			
$c = 3.00 \times 10^8 \ \frac{\text{m}}{\text{s}}$	$R = 8.314 \ \frac{\text{J}}{\text{mol} \cdot \text{K}}$	$M_{air} = 28.8 \ \frac{\text{g}}{\text{mol}}$	$g = 9.81 \ \frac{\text{m}}{\text{s}^2}$
$v_{sound\ at\ 20℃} = 343 \ \frac{\text{m}}{\text{s}}$	$T_K = T_{℃} + 273.15$	$\gamma_{monoatomic} = \frac{5}{3}$	
$I_0 = 1 \times 10^{-12} \ \frac{\text{W}}{\text{m}^2}$	$1\ \text{atm} = 1.01 \times 10^5\ \text{Pa}$	$\gamma_{diaatomic} = \frac{7}{5}$	
MATHEMATICAL FORMULAE			
$\cos(\alpha) \pm \cos(\beta) = 2\cos\left(\frac{1}{2}(\alpha \pm \beta)\right)\cos\left(\frac{1}{2}(\alpha \mp \beta)\right)$			
$\sin(\alpha) \pm \sin(\beta) = 2\sin\left(\frac{\alpha \pm \beta}{2}\right)\cos\left(\frac{\alpha \mp \beta}{2}\right)$			
$\cos(\alpha \pm \beta) = \cos(\alpha)\cos(\beta) \mp \sin(\alpha)\sin(\beta)$		$\tilde{A} = Ae^{j\theta} = A(\cos(\theta) + j\sin(\theta))$	
$\sin(\alpha \pm \beta) = \sin(\alpha)\cos(\beta) \pm \cos(\alpha)\sin(\beta)$		$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$	
$\sin(-\alpha) = -\sin(\alpha)$		$\cos\left(\alpha \pm \frac{\pi}{2}\right) = \mp\sin(\alpha)$	
$\cos(-\alpha) = \cos(\alpha)$		$\sin\left(\alpha \pm \frac{\pi}{2}\right) = \pm\cos(\alpha)$	